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LastMile-DDx: Explainable Home and Last-Mile Connectivity Diagnosis

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Problem Description

Home internet faults are diagnosed by guesswork; LastMile-DDx diagnoses them with evidence.

Speed tests are blind

They report peak bandwidth on an idle line only. Packet loss, RTT and loaded jitter, which decide call and video quality, go unmeasured.

Faults are hard to localise

Gateway, DNS, random loss and congestion all show up as "the internet is slow", so users cannot tell what to fix or whom to call.

Tools over-claim

Verdicts arrive with no evidence, or blame cables and ISPs nobody can verify, so users cannot trust or challenge them.

Our answer: LastMile-DDx

- Diagnosis by hypothesis elimination, like a clinician: six causes start alive and evidence rules them out
- Every verdict shows its rule, numbers and thresholds
- Media-aware thresholds for fiber, cellular, fixed wireless and satellite
- Returns Inconclusive plus an escalation report when tests cannot isolate the cause

Understanding the Problem Statement

Each requirement maps to one LastMile-DDx component, built out on the slides that follow.

Problem-statement requirement	How LastMile-DDx answers it
Diagnose slowness, not just measure speed	Probe gateway, public IP and DNS under idle and loaded conditions, after calibrating the host
Localise the fault to one cause	Six candidate hypotheses start alive; rules R1 to R5 prune them using measured evidence
Work across many access media	Media-aware thresholds for fiber, cellular, fixed wireless and satellite
Explainable, auditable verdicts	Each verdict prints the rule fired, numbers, thresholds, what was ruled out and the next test
Admit uncertainty honestly	Explicit Inconclusive verdict plus an escalation PDF, never a forced guess
No over-claiming without proof	No cable-damage, ISP-internal or Wi-Fi interference claims without radio telemetry
Prove it against known faults	Reversible fault lab: 30+ runs per fault, no-fault baseline and a confusion matrix

Background

One service, four access media: each fails differently, so one speed number cannot explain slowness.

Fiber

Low, stable RTT; most faults are local (Wi-Fi, gateway, DNS)

Cellular

Variable RTT; signal strength and cell load swing results

Fixed wireless

Sensitive to line of sight and weather; rain can add loss

Satellite

High RTT by design (GEO ≈ 600 ms), so it is not a fault

A speed test reports peak download on an idle line. Real-time quality depends on four things it never measures, and we measure each:

- **Packet loss**
- **Latency (RTT)**
- **Loaded jitter**
- **DNS delay**

Four Access Media: Key Differences

The same symptom can mean different things depending on how the home connects to the internet.

FIBER | Optical cable

Data travels as light through fiber. Usually low and stable RTT; local router, Wi-Fi, DNS or congestion may still cause problems.

CELLULAR | 4G / 5G

Mobile radio connects to a cell tower. Signal strength, mobility and tower load can change RTT, packet loss and throughput.

FIXED WIRELESS | FWA

A fixed receiver communicates by radio with a provider site. Line of sight, obstacles, antenna alignment, interference and capacity affect quality.

SATELLITE | GEO / LEO

Data travels via satellite. GEO has high propagation delay by design; LEO usually has lower delay. Obstructions and weather can affect signal quality.

Use media-aware baselines: high GEO satellite RTT alone does not prove a fault.

Network Metrics (1): Packet Loss and RTT

These measurements describe reliability and response time; they are not the same as download speed.

1. PACKET LOSS | Missing packets

Percentage of sent probe packets that do not return. Formula: lost packets ÷ sent packets × 100. Example: 5 missing out of 100 = 5% loss.

What packet loss affects

Voice and video can break up; games may lag; reliable transports may retransmit data. Causes include weak radio signal, congestion, faulty links or equipment.

2. RTT | Round-trip time

Time for a probe to reach a destination and for its reply to return, measured in milliseconds (ms). Example: 20 ms out + 20 ms back = 40 ms RTT.

How LastMile-DDx uses them

Probe the gateway and a public IP. Gateway loss suggests a local link/gateway issue; a clean idle path with RTT rising sharply only under load suggests congestion.

Interpret repeated probes against the connection type's baseline.

Network Metrics (2): Loaded Jitter and DNS Delay

The project checks both the network path under load and the service that translates website names.

3. LOADED JITTER | Delay variation

Jitter is variation in packet delay. Loaded jitter measures that variation while traffic is using the connection, such as during a download or upload.

Why loaded jitter matters

High or unstable jitter makes calls and live video choppy. Queues filling under load can create delay variation; radio conditions can also vary.

4. DNS DELAY | Name lookup time

DNS delay is the time needed to translate a domain name into an IP address. A slow or failing resolver can delay website opening even when the link works.

How LastMile-DDx tests them

Compare idle and loaded RTT/jitter. Test system DNS and an alternate resolver: if IP connectivity works, system DNS fails and alternate DNS succeeds, rule R2 points to a DNS-path fault.

Do not blame DNS from one failed lookup; compare resolvers and confirm IP reachability.

Literature and RFC Review

Standards and studies we build on, and how each one shapes LastMile-DDx.

Source / reference	Key idea from the literature / standard	How we use it in LastMile-DDx
MacMillan et al. (2023), A Comparative Analysis of Ookla Speedtest and NDT7	Speed-test tools can produce different results due to differences in design and infrastructure.	Motivates consistent probing and careful interpretation of measurements.
Clark & Wedeman (2024), Measurement of Internet Access Latency: A Cross-Dataset Comparison	Compares latency and jitter measurements across multiple platforms.	Supports RTT and jitter measurements and awareness of measurement limitations.
Ali, Hong & Cheung (2024), Congestion or No Congestion	Investigates how to distinguish congestion-related packet loss from wireless-link loss.	Supports differentiating random loss from congestion using measured evidence
RFC 6349 [6]	Framework for TCP throughput testing and interpreting performance under load.	Supports the loaded-path phase, comparing network performance under traffic load.
CP BBR-n Interplay with Modern AQM in Wireless-N/AC Networks (2024)]	Evaluates throughput and latency under loaded network conditions.	Supports our idle-versus-loaded network testing approach.
Gap addressed by our work	Many tools show a result without a transparent explanation or	The rule engine records evidence, shows fired rules and returns Inconclusive when evidence cannot isolate a cause.

Proposed Solution: Evidence-Based Diagnosis

Calibrated probes and explainable rules help users and ISPs verify the diagnosis.

Host calibration

Checks CPU load, timer accuracy and loopback behaviour before testing, reducing the chance that device-side errors distort the result.

Network probing

Measures gateway and public-IP reachability, DNS response, packet loss and RTT in idle and loaded conditions.

Candidate causes

Considers gateway failure, DNS fault, random loss, congestion, local Wi-Fi link issues and an unknown upstream fault.

Rule-based diagnosis

Applies transparent rules to the recorded measurements and removes hypotheses that do not match the observed evidence.

Explainable report

Presents the verdict, measurements, thresholds, ruled-out causes and next steps. If evidence is insufficient, the outcome is “Inconclusive.”

Guardrail: do not claim cable damage, ISP-internal faults or Wi-Fi interference unless supporting telemetry is available.

Explainable Decision Rules

Five ordered rules turn probe evidence into a verdict, and every verdict shows its working.

Rule	Evidence observed	Verdict
R1	Gateway ping 100% loss and ARP fails	Gateway unreachable: router or local link down
R2	Gateway OK, public IP OK, system DNS fails, alternate DNS OK	DNS failure on the local resolver path
R3	Idle loss above media threshold, RTT stable under load	Random packet loss, not congestion
R4	Idle path clean, loaded RTT above 3x baseline	Congestion (bufferbloat signature)
R5	Gateway OK, public IP failing, DNS failing	Inconclusive: escalate with evidence report

Every verdict prints the rule fired, the numbers and thresholds used, what was ruled out, and the next test that would change the answer, so users and ISPs can verify it.

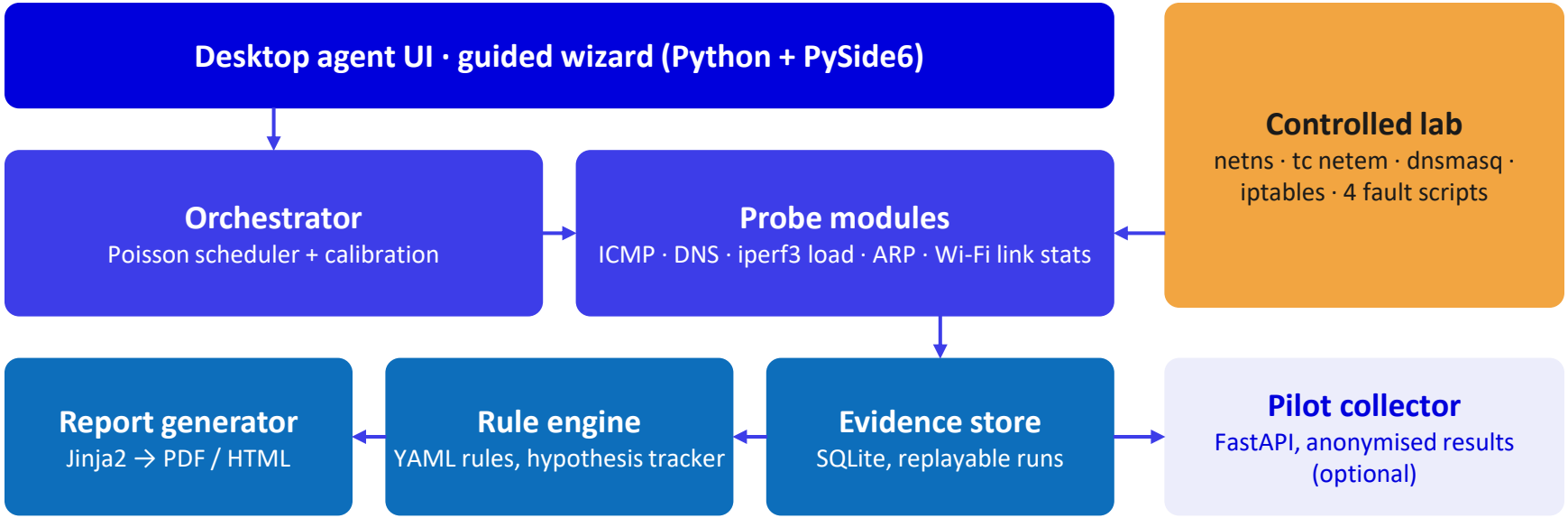
Optimization Proposed: Before and After

How LastMile-DDx improves on speed tests and opaque if/else diagnosers.

Aspect	Typical tool	LastMile-DDx
Decision logic	Opaque or plain if/else	YAML rules with an eliminated-hypothesis trace
Congestion vs loss	Often confused	Idle vs loaded RTT inflation test separates them
Ambiguity	Forced guess	Explicit Inconclusive plus the next test to run
Thresholds	One size fits all	Media-aware (fiber, cellular, FWA, satellite)
Measurement error	Ignored	Self-calibration of host and timer
Output	Single verdict	Replayable evidence and escalation PDF

Solution Architecture (UML Component View)

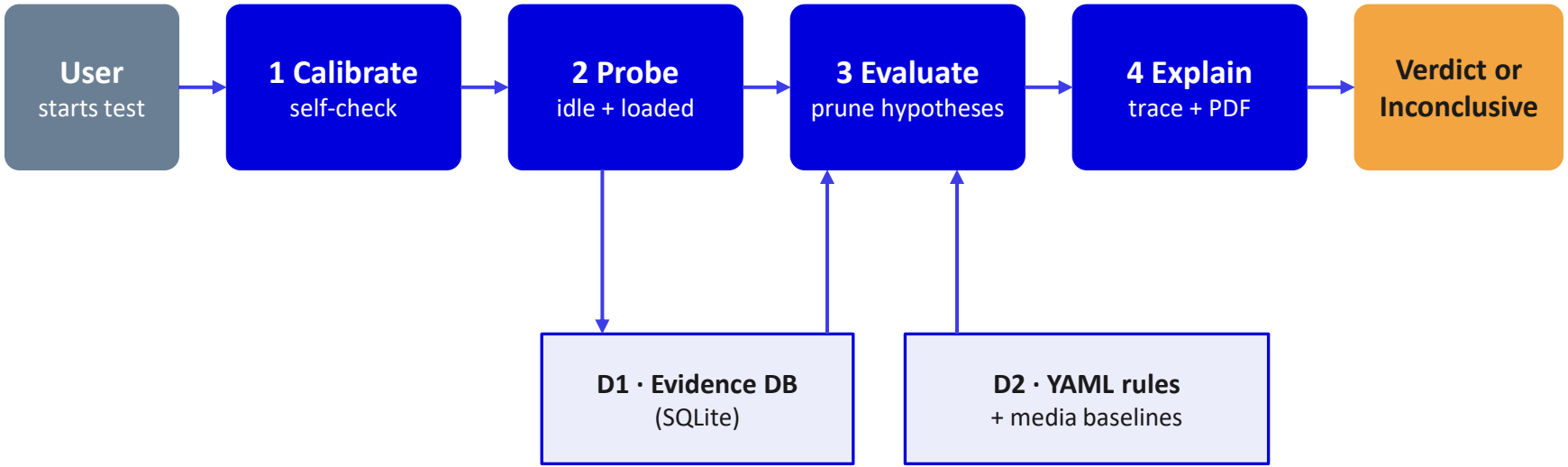
Eight components: probes in, evidence stored, rules applied, report out.



The rule engine reads only stored evidence, so every diagnosis can be replayed and audited. The pilot collector is optional and receives anonymised results only.

Data Flow Diagram

From the user's click to a verdict: evidence is written once, rules are read-only.



Probe results are written once to D1; the evaluator reads D1 and D2, so any verdict can be regenerated.

Rounded box = process · Open box = data store · Amber = output

Evaluation and Exposure Report

Evidence summary

Records gateway/public-IP reachability, packet loss, RTT, loaded RTT, jitter and DNS response.

Rule trace

Names the rule fired, thresholds applied, hypotheses eliminated and evidence supporting the verdict.

Evaluation results

Run 30+ trials per fault; compare predicted and injected faults using a confusion matrix, precision, recall and median diagnosis time.

Escalation report

If the result is Inconclusive, generate a PDF with test time, network type, measurements, ruled-out causes and recommended next test for the ISP.

The report is evidence-led: it avoids forced guesses and gives the user a clear, auditable next step.

Fault Lab and Evaluation Plan

Every rule is tested by injecting a known fault and checking that the verdict matches.

Fault	Injection (reversible script)	Expected verdict
Gateway unreachable	iptables DROP or veth link down	R1: gateway unreachable
DNS failure	Stop dnsmasq or force SERVFAIL	R2: DNS failure
Induced packet loss	tc netem loss 8%	R3: random loss
Network congestion	tbf rate limit + saturating iperf3	R4: congestion
Combined (loss + DNS)	Two fault scripts together	Inconclusive, by design (R5)

30+ runs
per fault scenario

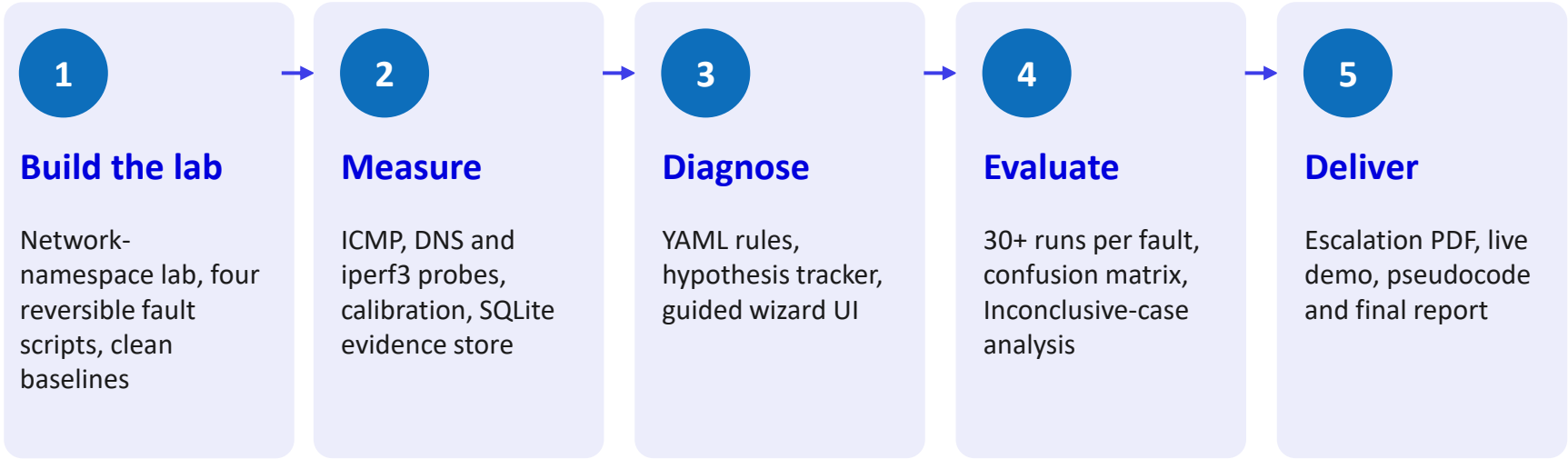
Baseline
no-fault false positives

Confusion matrix
precision and recall

Time to diagnosis
median per scenario

Deployment Timeline

Five sequential phases take us from an empty lab to a live, demo-ready system.



Each phase is checked against the fault lab before the next begins, and every run stores its configuration so it can be replayed. The complete system is demo-ready before the hackathon deadline.

Bibliography

[1] V. Paxson et al., “Framework for IP Performance Metrics,” RFC 2330, IETF, May 1998.

[2] G. Almes, S. Kalidindi, M. Zekauskas, “A One-way Delay Metric for IPPM,” RFC 2679, Sept. 1999.

[3] G. Almes, S. Kalidindi, M. Zekauskas, “A One-way Packet Loss Metric for IPPM,” RFC 2680, Sept. 1999.

[4] G. Almes, S. Kalidindi, M. Zekauskas, “A Round-trip Delay Metric for IPPM,” RFC 2681, Sept. 1999.

[5] P. Eardley et al., “A Framework for Large-Scale Measurement of Broadband Performance (LMAP),” RFC 7594, Sept. 2015.

[6] B. Constantine et al., “Framework for TCP Throughput Testing,” RFC 6349, Aug. 2011.

[7] J. Gettys and K. Nichols, “Bufferbloat: Dark Buffers in the Internet,” ACM Queue, vol. 9, no. 11, 2011.

Thank You for Your Attention!

Diagnose with evidence. Escalate with confidence.

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